QUESTION 1

What is the primary difference between a microcontroller and a microprocessor ?

Answer is Wrong **Score: 0 / 1**

Options

A. Microprocessors have more peripherals integrated than microcontrollers.

Expected

B. Microcontrollers integrate CPU, memory, and peripherals on a single chip, while microprocessors require external support circuitry.

C. Microcontrollers cannot perform computations.

Wrong

D. Microcontrollers cannot perform computations.

FEEDBACK

Solution :Microcontrollers combine CPU, memory, and peripherals on one chip, whereas microprocessors need external components for functionality. Youtube link : https://youtu.be/SQJYqcqAg_w?feature=shared&t=270

QUESTION 2

Which of the following is/are advantages of using microcontrollers?

Not Attempted **Score: 0 / 1**

Options

Expected

A. Reduced power consumption due to integrated components.

B. Higher computational power than microprocessors.

Expected

C. Lower cost and simplified inventory management.

D. Ability to handle complex video processing tasks efficiently.

FEEDBACK

Solution :Microcontrollers reduce power and cost by integrating components but sacrifice raw computational power for specialized tasks. Youtube link : https://youtu.be/SQJYqcqAg_w?feature=shared

QUESTION 3

What is the role of ROM in a microcontroller?

Not Attempted **Score: 0 / 1**

Options

A. To store temporary variables during computation.

Expected

B. To hold pre-programmed configuration data and boot instructions.

C. To act as high-speed secondary memory.

D. To manage peripheral communication.

FEEDBACK

Solution :ROM stores non-volatile data like boot instructions and configuration, unlike RAM for temporary variables. Youtube link: https://youtu.be/SQJYqcqAg_w?feature=shared&t=380

QUESTION 4

Why might a designer choose a microprocessor over a microcontroller?

Not Attempted **Score: 0 / 1**

Options

A. To integrate peripherals like USB and Bluetooth.

Expected

B. To prioritize high computational power and expandability.

C. To minimize the number of external components.

D. To reduce standby power consumption.

FEEDBACK

Solution :Microprocessors are preferred for tasks requiring high-speed computation and flexibility, outsourcing peripherals/memory. Youtube link: https://youtu.be/SQJYqcqAg_w?feature=shared&t=725

QUESTION 5

Which of the following are typically integrated into a microcontroller?

Not Attempted **Score: 0 / 1**

Options

Expected

A. UART (Universal Asynchronous Receiver Transmitter)

B. Lcd display

Expected

C. Timers and general-purpose I/O

D. Secondary storage (e.g., HDD)

FEEDBACK

Solution :Microcontrollers often include UART, timers, and general purpose I/O. Youtube link: https://youtu.be/SQJYqcqAg_w?feature=shared&t=470

QUESTION 6

How has the distinction between embedded systems and general-purpose systems blurred?

Not Attempted **Score: 0 / 1**

Options

A. Microcontrollers now outperform microprocessors.

Expected

B. Appliances like refrigerators incorporate touchscreens and Wi-Fi, resembling general-purpose devices.

C. Embedded systems no longer use microcontrollers.

D. General-purpose systems avoid peripheral integration.

FEEDBACK

Solution :Devices like smart refrigerators blend fixed functionality with expandable features, blurring traditional categories. Youtube link: https://youtu.be/SQJYqcqAg_w?feature=shared&t=961